Nadeeshan De Silva

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Objective

I am a 3rd year PhD student in Computer Science at William & Mary, with 2 years of industry experience as a software engineer and over a year of research as a Research Assistant. My research focuses on Software Engineering, Machine Learning, and Natural Language Processing, with a strong interest in applying ML to improve developer tools.

Education

The College of William and Mary, Williamsburg, VA — Aug 2022 - present
Ph.D. in Computer Science (GPA 4.00/4.00) | Advised by: Dr. Oscar Chaparro

University of Moratuwa, Sri Lanka — Feb 2016 - Jan 2020
B.Sc. Engineering (Hons) in Computer Science & Engineering (GPA 3.53/4.20)

Publications

- J Mahmud, **N De Silva**, S A Khan, S H Mostafa, S M Hasan Mansur, O Chaparro, A Marcus, and K Moran "On Using GUI Interaction Data to Improve Text Retrieval-based Bug Localization" accepted to *International Conference of Software Engineering*, ICSE 2024 [arxiv] (acceptance rate: ~22%)
- Y Song, J Mahmud, **N De Silva**, Y Zhou, O Chaparro, K Moran, A Marcus, and D Poshyvanyk "BURT: A Chatbot for Interactive Bug Reporting" *International Conference of Software Engineering*, ICSE 2023 tool demo track [IEEE Xplore] (acceptance rate: ~26%)
- G Gamage, **N De Silva**, I Perera, S Bandara, T Pathirana, A Wickramarachchi, and V Mallawaarachchi "Phylogenetic Tree Construction Using K-Mer Forest-Based Distance Calculation" in *International Journal of Online & Biomedical Engineering*, IJOE 2020. [IJOE].
- (**Best Paper Award**) G Gamage, **N De Silva**, A Wickramarachchi, V Mallawaarachchi and I Perera "Alignment-free Whole Genome Comparison Using k-mer Forests" in *19th International Conference on Advances in ICT for Emerging Regions,* ICTer 2019 [IEEE Xplore] (acceptance rate ~23%)
- N Bandara, **N De Silva,** H Guruge "Increasing the Quality of Patient Care through Stream Processing" – White Paper 2019 In [InfoQ]

Industrial Experience

Graduate Research Assistant, William & Mary, VA, USA — August 2024- present
- Conducting machine learning-based experiments to investigate the effect of relative comprehensibility as a new metric to improve automatic code comprehensibility.
- Investigated the effect of mobile GUI interaction data to improve text-retrieval-based bug localization.

Senior Software Engineer, Enactor, Sri Lanka (Branch of Enactor Limited, Hertford, UK) — Feb 2020 - July 2022
- **Engineered** a customized **React-based POS module** for an enterprise-level client, delivering a more efficient and user-friendly interface.
- **Enhanced platform code quality by 20%** through systematic refactoring, resolving over 100 **SonarQube-reported issues**, leading to cleaner, maintainable, and efficient code.
- Engaged in various design, development, deployment, quality assurance, and customer support tasks.

Google Summer of Code Intern (Open-Source Project), The Apache Software Foundation, Remote — May 2019 – _Aug 2019
- Designed and implemented modularized, reusable React components and Improved JAX-RS-REST APIs for Apache OODT 2.0. [project]

Software Engineer Intern, WSO2, Sri Lanka — June 2018 - Dec 2018
- Developed a prototype for a Healthcare Data manipulation that supports global HL7/FHIR standards as my internship project. (Kafka/Zookeeper mechanism, stream processing-based data processing) and published a white paper in InfoQ [paper]

Skills

- Programming Languages: Java, Python, JavaScript, Bash, and CUDA.
- Operating Systems: Unix/Linux, MacOS, and Windows.
- Other: PyTorch, NumPy, Pandas, Scikit-learn, Docker, SQL, React, Android, OpenCV

Achievements

- Received International Student Opportunity Scholarship from William and Mary, VA, USA — 2024
- Received ACM SIGSOFT travel grant to attend the International Conference on Software Engineering (ICSE). — 2024
- Recognized for academic excellence on graduating with SGPA 3.98 /4.2 with an A+ for the bachelor's Thesis project. — 2020
- The Best Paper Award at the International Conference on Advances in ICT for Emerging Regions(ICTer) — 2019
- World Rank 330/7000+ - Deng AI-Driven Data competition — 2019
- Sri Lankan Finalists – TAD Hack (Computer Vision based application to convert handwriting to PDF near real-time) — 2017
- 4th place in the Hackathon – HACKLN — 2017

Professional Qualifications/Activities

- ESSEC (Eastern Atlantic Students in Software Engineering Colloquium), William & Mary, Williamsburg, VA Jan 2024
- ICSME (The International Conference on Software Maintenance and Evolution) Student Volunteer Oct 2023
- Mentor and Judge @ Cypher VIII Hackathon at William and Mary, Williamsburg, VA Oct 2022, Nov 2023
- Appointed as Apache OODT Project Management Committee (PMC) Member and Committer. April 2021